

PHYS 230 - Electricity and Magnetism

Lecture I

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

Course Information

1 Electric Charge and Electric Field

- Electric Charge
- Conductors, Insulators and Induced Charge

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Course Information

- **Instructor:** Saeed Rastgoo
- **Office:** CCIS 2-095
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- **Office hours:** See eClass “[Announcements](#)”
- **Extra office hours:** only by appointment (send request via email)
- **Further details:** see [Course Syllabus](#)
- **Polls:** To make problem solving interactive in class, we will often use ePoll
 - Polls are anonymous and not graded, it's only for learning
 - The link is: <https://epoll.srv.ualberta.ca/> and the code is MBD

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- The **SI units of electric charge** is **coulomb** denoted by **C**

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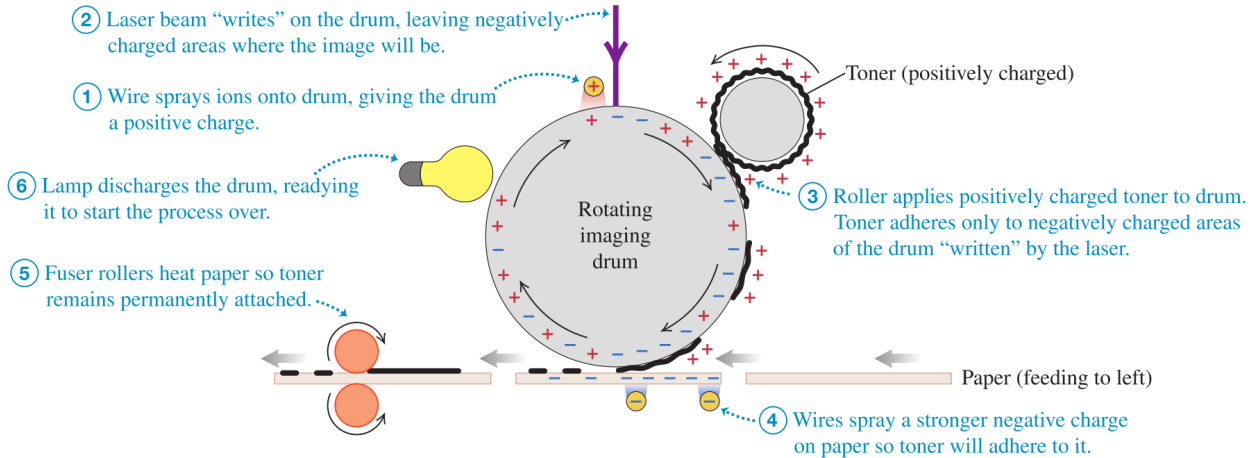
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- Any observer (including another charge) **not at rest** w.r.t. a charge q sees **magnetism** in addition to electricity from q

Example of Use of Electric Charge Properties

Figure 21.2 Schematic diagram of the operation of a laser printer.



Structure of Matter

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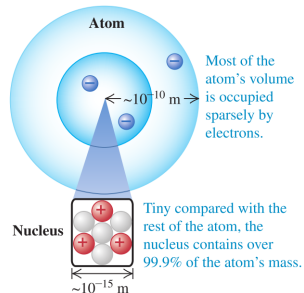
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Figure 21.3 The structure of an atom. The particular atom depicted here is lithium (see Fig. 21.4a).



- **Proton:** Positive charge
Mass = 1.673×10^{-27} kg
- **Neutron:** No charge
Mass = 1.675×10^{-27} kg
- **Electron:** Negative charge
Mass = 9.109×10^{-31} kg

The charges of the electron and proton are equal in magnitude.

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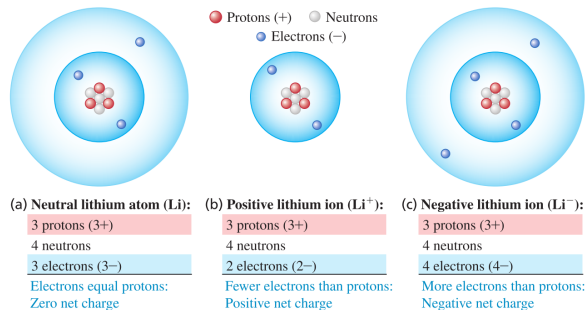
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Conservation and Quantization of Charge

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The net electric charges in any closed system is constant

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QUANTIZATION OF ELECTRIC CHARGE

The magnitude of charge of the electron or proton is a natural unit of charge

Every observable amount of electric charge is always an integer multiple of this basic unit

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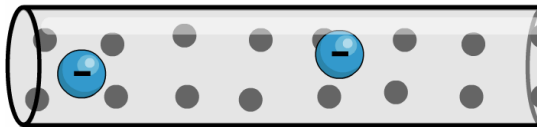
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Conductors and Insulators

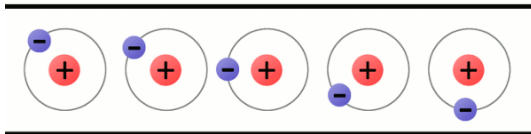
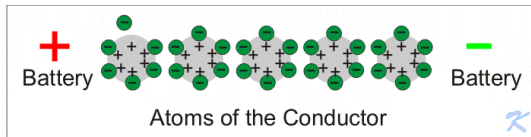
Conductors: Materials that permit electric charge to move through them; Have loose/free electrons in their outer atom layer

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Time: 0 seconds

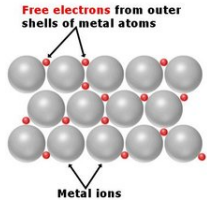


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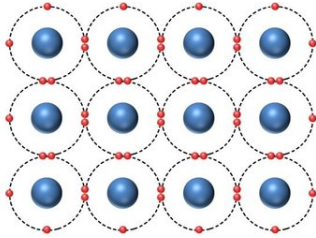
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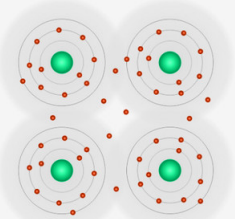


Conductor

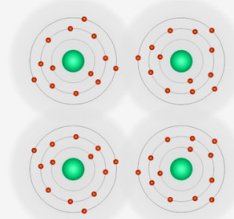


Insulator

conductor structure



insulator structure

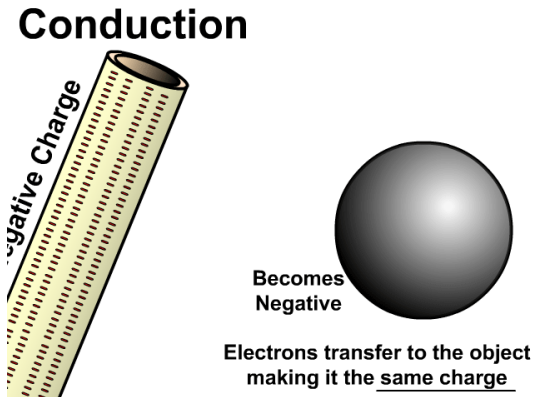


Charging by Conduction

- Charging by **conduction**: Direct transfer of charge from one object to another by direct contact

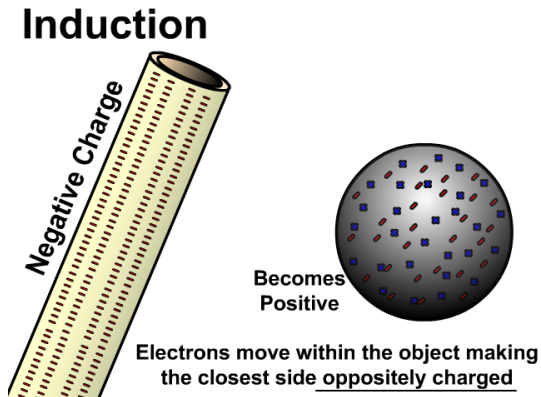
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- Charging by **conduction**: **Direct transfer of charge** from one object to another by **direct contact**
- One or both object will have excess of one type of charge



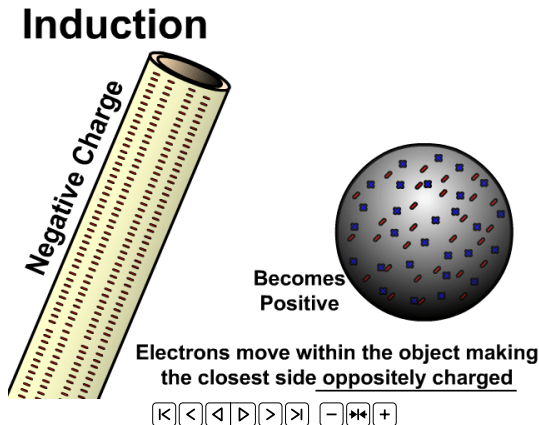
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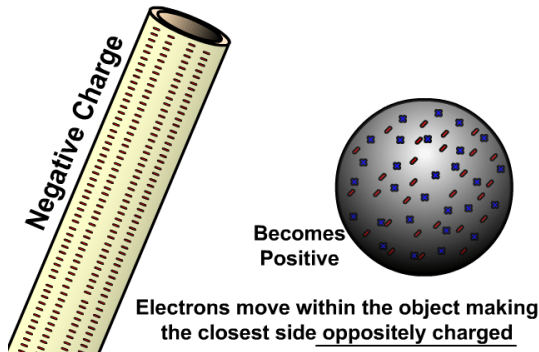
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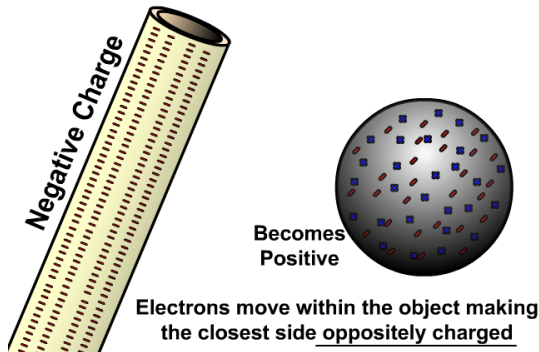
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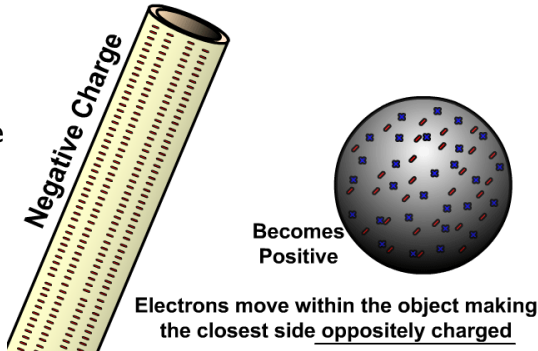
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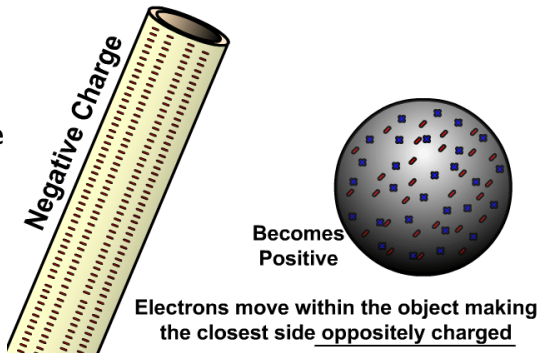
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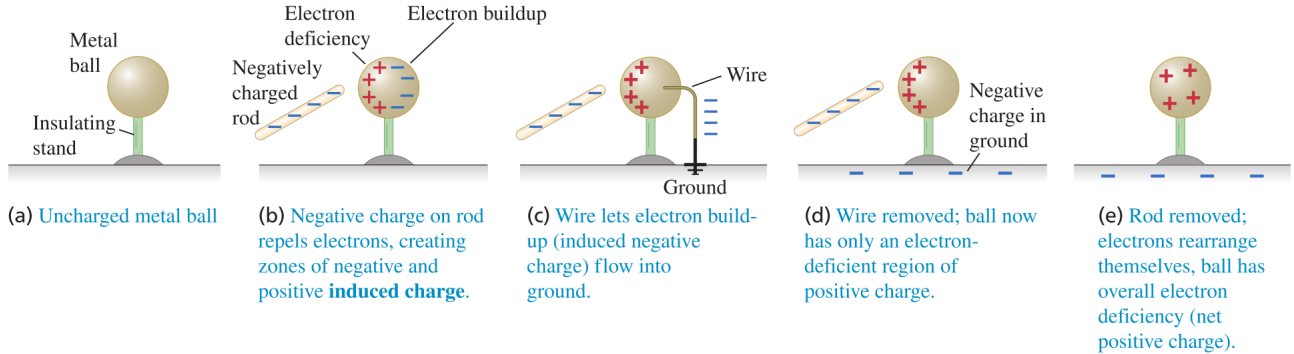
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- These excess charges are called induced charges

Induction



Combination of Induction and Conduction

Figure 21.7 Charging a metal ball by induction.



Attraction in Insulators

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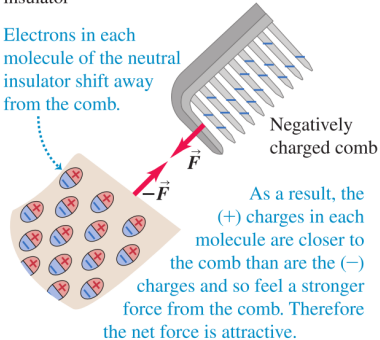
Figure 21.8 The charges within the molecules of an insulating material can shift slightly. As a result, a comb with either sign of charge attracts a neutral insulator. By Newton's third law the neutral insulator exerts an equal-magnitude attractive force on the comb.

(a) A charged comb picking up uncharged pieces of plastic



(b) How a negatively charged comb attracts an insulator

Electrons in each molecule of the neutral insulator shift away from the comb.



(c) How a positively charged comb attracts an insulator

This time, electrons in the molecules shift toward the comb ...

